Against the Frontier: A Cross-Engine Measurement and the Bridge Defect That Nearly Buried It

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Abstract

Every result this project had published about its own strength was measured inside its own engine, against its own bots. This paper reports the first measurement against an independently developed frontier agent — SESTINA v1.0, the strongest bot of the FishLab project [3] — played inside *their* C++ engine, over *their* documented guest-bot protocol, on 7,200 games across six seeds. **FishAI wins** 27.08%**.** It sits below their entire published lineage, losing to v0.5 and v0.6 as well, and its placement is between their v0.3 and v0.4 — the release that added an exact deal posterior, four releases before search existed in that lineage. The anatomy is not the one the first version of this measurement reported. FishAI declares as accurately as the frontier does (98.42% against 98.46%); what it cannot do is *prove* what its team already owns, holding a resolved set for 9.30 events where SESTINA holds it for 2.92. A cheating oracle that cashes every lock instantly is worth +5.75 points of the 22.17 to even — 25.9% of the gap, and a ceiling rather than a plan. The remaining ∼16.4 points are unattributed and this paper does not claim to have measured them. The methodological half of the paper is a defect: the first version published 24.22%, a figure depressed 3.44 points by a rules-dialect mismatch in the bridge, which survived two adversarial audits, eight zeroed corruption counters, and a mirror-cell control that returned a perfect 50.0000% across a four-point hole because a mirror cell has, by construction, zero statistical power. What caught it was a protocol counter reading zero where zero was impossible. We report the defect, the withdrawn numbers, and three instrument rules that follow.

1 Introduction

A laboratory that only ever plays itself learns the shape of its own roster. The FishAI project [6, 9] is an engine, bot laboratory and results site for the six-player partnership card game Literature (Canadian Fish) in its 54-card `us54` dialect [5], and until this measurement every claim it had made about strength was internal: nine styles sharing one inference engine, played against each other on duplicate deals. That design is the right one for the question it was built to ask — *is any style superior?* — and it is the wrong one for the question a reader actually wants answered, which is whether any of it is any good.

The FishLab project [3] supplies an answer to that second question, because it is an independently written Canadian Fish engine with its own bot lineage running v0.2 through v0.6 and a frontier agent, SESTINA v1.0, which is that lineage's v0.7. Its rules agree with `us54` exactly on every row that matters: 54 cards, nine half-suits of six, seats 0–5 with teams by parity, an ask requiring the asker to hold a card of the named set, a hit keeping the turn, a miss passing it to the seat asked, and a wrong declaration awarding the set to the opponents. Its C++ engine also carries out-of-turn declaration, which `us54` row 11 has and their TypeScript lab does not.

This paper reports what happened when FishAI's decision function was registered as a guest bot in that engine and played against SESTINA v1.0. It has two halves and they are of roughly equal weight. The first is the standing and its anatomy (Sections 3–5). The second is a defect (Section 6), which is in the paper at all because the first version of this measurement published a number that was wrong by 3.44 points, and every control in the setup said it was fine.

Scope, stated once and meant. Nothing here is a claim about a *tuned* FishAI: no FishAI parameter was fitted against SESTINA, and doing so would burn it as a holdout. Nothing here is a claim about the adaptive arm [7], which the adapter is known to be unsafe for (Section 7). And nothing here is a re-measurement under the `us54` turn-pass correction of September 2026 [5]: the games below were played by the policy as it stood before that correction, and while the host engine's turn flow — not FishAI's — governed every game reported here, the policy that played them was tuned inside the uncorrected engine. Re-running these cells against the corrected policy is future work and is named as such in Section 8.

2 The harness

2.1 What runs, and where

SESTINA v1.0 is a frozen policy *specification*, not a binary: the release ships `fishbot_v07.json`, a paper and its sources, and is played by handing a spec string to their C++ engine [3]. The opponent in every cell below was verified byte-identical in its `spec` and `allparamsSpec` fields against that release asset before a game was played [1].

Their engine builds from source in a Linux container — `node:24-bookworm` plus `g++`, `make` and `zip`. Their Makefile defaults to `clang++`; g++ 12 builds it unmodified in about 35 seconds. Node is needed in the *same* image, because the adapter is a Node process their engine spawns as a guest seat, and FishAI's TypeScript needs no build step there: Node 24 imports the engine directly by type-stripping.

One build flag is load-bearing. Their Makefile defaults to `-march=native`, which enables FMA, changes floating-point tie-breaks, and makes their own published identity digests irreproducible across machines. Building with `-march=native -ffp-contract=off` reproduces the generic digest exactly, and all three of their identity controls **PASS** on that build. Every cell reported here is the portable `-ffp-contract=off` build [1].

2.2 The bridge

FishAI plays through an adapter that translates their game state into a FishAI `SeatView` and translates FishAI's `GameAction` back, over their documented `fishlab-json-v1` line-delimited protocol [3]. The adapter is original work written from their published protocol document; no code or prose of theirs is copied into this project, their repository carries no licence file, and their measured findings are cited as theirs throughout.

Two asymmetries in the bridge run in FishAI's disfavour and are recorded because a bridge that quietly helps the guest measures nothing:

- **The failed-declare channel is strictly weaker than at home.** On a failed declaration their protocol reveals nothing about true holders, while FishAI's internal event carries them. The adapter emits only cards already public from an earlier hit, so FishAI sees *less* than its

Table 1: The governing paired floor by sample size [2]. Single-cell differences smaller than ∼6.9 points are not resolvable, which is why no conclusion below rests on one cell.

sample	deals	paired floor
one cell	200	±6.93
three seeds	600	±4.00
five seeds	1,000	±3.10
six seeds	1,200	±2.83

home engine would give it and never sees anything false. Strictly weaker inference, never wrong inference; it cannot inflate the result [1].

- **Every corruption counter read zero** across the whole measurement — `planMismatch`, `booksDisagree`, `successHolderClash`, `viewInvariant`, `declareShapeBad`, `traceFallback`, `askNotAsk`, `forcedOwnTeamOut` — and Section 6 is about a run that was defective anyway. Corruption counters detect corruption, not mistranslation.

2.3 The cell, and the floor it is read against

A cell is **200 deals × 6 rotations = 1,200 games**, duplicate-dealt: every deal is replayed with the seats rotated. Zero faults and zero action-limit games in every cell.

Their engine prints two power figures on every run and **the per-deal one governs**: about ±2.83 points unpaired over 1,200 games, against ±6.93 points over 200 deals, which is the paired floor. The 1,200 games of a cell are not independent draws — they are 200 deals seen six ways — so a Wilson interval computed on the game count runs about 2.4× too tight. Every interval in this paper is per deal.

3 Standing

3.1 The headline

FishAI wins 27.08% **of games against SESTINA v1.0** — 1,950 of 7,200 games over six seeds [1].

That is 22.92 points under even, and it is not a surprise in its *direction*. SESTINA v1.0 is a determinization search agent: it samples worlds from a particle belief and searches each one. FishAI's arm is `STYLE_ROSTER.punter` at full-strength inference, a heuristic policy that scores the legal moves once. Their own materials price SESTINA at at least 4.52× the cost of the non-searching blueprint it descends from [3]. A policy that scores moves once holding parity against a search agent spending four and a half times the compute would have been the surprise.

The size is the finding, and so is the company it keeps.

3.2 Below the whole lineage

Table 2 is the result that hurts. FishAI does not merely lose to the frontier; it loses to *four* of their six published bots, including two that predate search entirely. It is furthest under even against SESTINA — 5.03 points below its v0.6 cell and 5.47 below its v0.5 cell, both clearing the ±4.00 paired floor *because the three opponents were played on the same three seeds.* Read cell-to-cell across different seeds nothing here separates them.

Table 2: FishAI against the full FishLab lineage, corrected bridge, six opponents on three matched seeds [2]. 600 deals per row, governing floor ±4.00.

FishAI vs	win rate	vs even	FishAI hold	opp. hold	FishAI dec. acc	opp. dec. acc
FishBot v0.2	**67.42**%	+17.42	6.67	19.84	98.90%	87.29%
FishBot v0.3	**62.06**%	+12.06	7.94	12.05	98.22%	85.62%
FishBot v0.4	**34.25**%	−15.75	8.02	4.87	97.69%	98.30%
FishBot v0.5	**33.31**%	−16.69	8.34	4.00	98.24%	97.55%
FishBot v0.6	**32.86**%	−17.14	8.34	3.71	97.53%	97.86%
SESTINA v1.0	**27.83**%	−22.17	9.24	2.97	98.37%	98.33%

Table 3: Per-decision detail, seed 90210, both bridges against the same SESTINA [1]. The defective column is reported so the correction in Section 6 can be read against it, not because it means anything on its own.

	FishAI (defective)	**FishAI (corrected)**	SESTINA v1.0
declare accuracy	92.83%	**98.42**%	**98.46**%
ask accuracy	51.95%	**52.32**%	**57.38**%
lock hold, events before cashing	9.55	**9.30**	**2.92**
declarations per game	3.74	3.64	5.31
out-of-turn declares per game	2.60	2.67	3.65

The placement is between v0.3 and v0.4, and their ladder is not a ladder. Running their own bots against each other through the same harness resolves only one of three steps:

anchor	win rate	resolved against even at ±4.00?
v0.3 vs v0.2	51.17%	no
v0.4 vs v0.3	**74.72**%	**yes** — +24.72
v0.5 vs v0.4	50.86%	no

Essentially all of their lineage's strength arrives in one step, v0.3 → v0.4, and FishAI sits on the near side of exactly that step. **v0.4 added the exact deal posterior and the locked-half-suit theorem** — four releases before search existed in that lineage [3]. This is cited as their release history, not measured here.

The harness reproduces their own results. Two anchor cells run their bots against each other: SESTINA beats v0.6 at 54.17% and v0.5 at 55.58%, against their published +4.63 pp and +5.18 pp over even, i.e. 54.63% and 55.18%. Both fall inside the measured cells [1, 3].

4 The anatomy: declaration is not the gap

FishAI declares as accurately as the frontier does. 98.42% against 98.46% is a gap of four hundredths of a point. The first version of this measurement named declaration quality as the largest genuine channel, on a 5.24-point deficit; that deficit was the bridge (Section 6) being read as a property of the policy, and it is withdrawn.

Two gaps survive the correction, and the defect barely touched either:

- **Asking:** 52.32% **against** 57.38%, a 5.06-point deficit — now the larger of the two behavioural gaps rather than the smaller.

Table 4: The oracle ablation on the corrected bridge [2]. Three seeds, 600 deals, floor ±4.00; per seed +4.17/ + 4.58/ + 8.50, 3/3 positive.

	base	oracle	delta
win rate	27.83%	**33.58%**	+**5.75**
lock hold	9.24	**0.41**	−8.83
declare accuracy	98.37%	99.64%	+1.27
ask accuracy	52.35%	56.10%	+3.75
declarations / game	3.650	3.810	+0.160
set differential	−1.573	−1.175	+0.398

- **Cashing:** 9.30 **events against** 2.92, a factor of 3.2. FishAI sits on a resolved set more than three times as long as SESTINA before declaring it, and a set held is a set that can still be taken apart. SESTINA declares 5.31 times a game to FishAI's 3.64: it is not declaring more *accurately*, it is declaring more *often and sooner*, at the same accuracy.

4.1 What the lag is, and what it is not

The obvious reading of a 9.30-event lock hold is a policy leak — a bot sitting on sets it could cash. **It is not.** Decomposing the hold over 1,970 declared locks against the ground-truth moment each half-suit became locked gives a mean HOLD of 8.97 events, of which LAG (first proof minus true lock) is 8.27 and POLICY (declare minus first proof) is **0.01** [2].

The load-bearing evidence for POLICY ≈0 does not depend on that decomposition. Across **265,884 declined declare polls** in two independent probes, *not one* was declined while a provable set sat in hand. **The metric measures how long FishAI is blind, not how long it waits.**

The confound, stated plainly. Lock hold is not a pure capability. A team that is winning acquires its proofs from the same events that win it cards — 72.5% of locks are proved at the instant of lock — so a short hold is partly a *consequence* of doing well. Table 2 is the direct evidence: one byte-identical FishAI arm moves 6.67 → 9.24 across six opponents with no code change, ordered by its own win rate in the cell. **Nothing in this paper separates "FishAI proves slowly, therefore it loses" from "FishAI loses, therefore it proves slowly."** No cross-cell comparison of lock hold is admissible; only the within-cell contrast is quoted, and even that ordering is consistent with the confound. *[inferred]*

5 How much of the gap is reachable

The one intervention in this dossier that bears on the direction of the arrow is a cheating oracle: the three FishAI seats share their true hands and cash every lock instantly.

+5.75 **points is the ceiling of the whole detection channel, and it is a cheat.** Three things follow and all three matter.

1. **Perfect lock detection closes** 25.9% **of the gap to even** — 5.75 of 22.17.
2. **The internal accounting agrees.** The oracle closes 0.398 of the 1.573-set differential, 25.3%, and $0.398 \times 14.7 = 5.85$ predicted points against +5.75 measured. Two routes agreeing inside 0.10 points.

3. **The count gap is not closable by proving faster.** The oracle buys only 0.160 declarations per game, 19.5% of the 1.638-declaration gap. The accounting identity that says "equalise declaration counts and the differential moves $-1.86 \to -0.50$, worth +20 points" is an identity, not a reachable target: declaring more sets requires *winning more cards*, not only proving faster.

The delta survived the correction, and that is the strongest single result here. The same oracle delta measured on the defective bridge was 30.17% against 24.42% — +5.75 to two decimals — while both of its *levels* moved about 3.4 points. Both arms carried the same defect, so it cancelled. An ablation can be sound while the arm it ablates is not.

The residual is the largest single term and nobody measured it. 22.92 points to even, of which the proof-latency ceiling is 5.75. **Something else owns roughly 16.4 points and nothing in this dossier measured what.** No term in this paper may be called dominant while that sentence stands. It is stated here rather than in a footnote because the temptation to present the largest *identified* term as the explanation is exactly the error this project has already made once.

Terms do not sum. Adding the measured channels is the specific error this project has documented before: two mechanisms worth +1.915 and +2.068 sets alone measure +1.565 *together* [10]. The cross-play re-measurement confirms it with an explicit factorial, in which two of its own large terms carry an interaction of −4.12 points [2]. Any ranking in this paper is a ranking, not a budget.

6 The defect

6.1 What it was

`us54` has no pass. When a seat holds the turn and has no cards, the rules compel it to declare on best evidence, because the alternative is a table that never progresses [5]. **Their engine does have a pass**, and sends it the moment the declare poll is answered "none".

The adapter translated FishAI's compulsion into a host that does not impose it. Polled at a cardless turn-holder, FishAI answered with a compelled, speculative declaration instead of declining and being handed the pass it was about to be offered. **Every such declaration spent a set to avoid a move that costs nothing.**

The fix is one guard in the adapter's declare-poll handler, deliberately narrow: it fires only when the hand is empty, the seat holds the turn, and the decision is one of the two *compelled* trace kinds. A confident own-book, certain or expected-value claim still goes through untouched, so the guard suppresses compulsion and nothing else. **The defect is in the bridge, not in FishAI**: the policy was answering a question the host never asked, and `us54` genuinely has no pass, so the empty-hand arm is correct for the dialect it implements.

6.2 What it cost

Mean delta +3.44 points, SD 0.48, smallest +2.67, and 5.59 points of declare accuracy. Pooled over 1,200 deals the paired floor is ±2.83, so the correction clears it — and *no single cell would have*, which is Section 2.3's point made in anger.

Table 5: Both arms on identical deals, six seeds [1]. Every seed moves the same way.

seed	defective	corrected	delta
90210	24.33%	28.25%	+3.92
4242	25.08%	28.83%	+3.75
7011001	23.25%	26.42%	+3.17
13579	22.83%	26.17%	+3.33
24680	23.42%	27.25%	+3.83
31415	22.92%	25.58%	+2.67
pooled	23.64%	**27.08%**	**+3.44**

6.3 Why nothing caught it

This is the part worth generalising, because the defect survived every check the setup had.

The mirror cell was not a control. The first version reported "FishAI vs FishAI, same harness: exactly 50.0000% $[47.18, 52.82]$" and read it as proof the bridge was symmetric. Their engine prints the reason it proves nothing on the mirror cell's own power line: the win-rate effective sample is zero, because the per-deal outcome is deterministic. A mirror cell plays one policy against itself on duplicate deals, every deal is replayed with the seats rotated, and the aggregate is **forced to** 50% **by construction** before a card is dealt. The published Wilson interval beside it was computed on a denominator the engine had already declared meaningless.

It is not merely uninformative — it is uninformative about exactly the defect that was present. A mirror cell cannot see a bug that affects both sides equally, and this bug affected both sides equally: it cost the defective arm 3.44 points against SESTINA while leaving the mirror at 50.0000%, because both mirrored seats lost the same sets in the same positions. **The control returned a perfect score across a four-point hole**, and no rearrangement of seats or seeds repairs that. It is a property of mirror controls, not of this run.

What did find it had already fired. The adapter counts every protocol operation it services. `opPass` read **0** across 3,600 games — a cardless turn-holder never once passed. The first version recorded that number and filed it as "unreached, not verified". It was not an unreached branch. It was the symptom, printed in the artifact, of the defect above.

Two independent adversarial audits returned sound-with-caveats, and both missed it. Record that as a fact about audits.

The lesson, narrowly. A control that cannot fail is not a control, and a counter reading zero where zero is impossible is not a coverage gap. The failure mode is specific to cross-engine work: a *rules-dialect mismatch* is what any bridge is most exposed to and least able to see, because both sides are internally consistent and only their seam is wrong.

6.4 Numbers withdrawn

Named rather than deleted, because a published number does not stop existing [1].

The four pooled counters are withdrawn on the authority of FishLab's own documentation, which states in two places that the pathology tool pools both sides and warns that a pooled figure in a

Table 6: The withdrawn figures from the first version of this measurement.

withdrawn	was reported as	status
24.22%	FishAI vs SESTINA v1.0	superseded by **27.08**%
50.0000% [47.18, 52.82]	"the control passed"	**void**; zero effective sample
93.25% declare accuracy	FishAI's declare rate	superseded by 98.42%, at parity
"the mechanism is declaration"	the anatomy	**reversed**; asking and cashing are the gaps
53.85% of forced declarations wrong	"the worst number in the run"	**withdrawn**; pooled over both teams
8.83% asks into own locked sets	a FishAI targeting bug	**withdrawn**; pooled
47.68% repeat (actor, suit, target)	a FishAI pathology	**withdrawn**; pooled
4.21% of declarations wrong	FishAI's base rate	**withdrawn**; pooled
per-cell CIs of $\sim\pm 2.4$ pts	precision of a cell	**too narrow**; the floor is ± 6.93

cross match is therefore unreliable [3]. **That caveat was in their repository throughout and was read past.**

7 Caveats that travel with the numbers

None of these is filed somewhere a number can travel without it.

- **Roster style was never varied.** The arm is Punter throughout, and no claim is made that Punter is the right arm abroad — only that it is the arm that ran. A style sweep on the *defective* bridge separated four roster styles by well under a per-cell floor and has not been repeated.

- **One `major` audit finding stands.** The observation layer mis-replays counts from the partial holders map. It is unreachable at a fixed roster style, so it does not touch this result — and it means **the adapter must not be reused for the adaptive arm without a fix**.

- **The seeds share a build and an adapter.** They are independent deals, not independent implementations, and Section 6 is what a shared adapter is worth as a risk.

- **The frontier is not claimed to be the frontier.** FishLab decline the strongest claim themselves, separating "strongest configuration this project has produced", which they record as established, from global standing, which they record as not established and not claimed. There is no published basis, theirs or this paper's, for treating SESTINA as the strongest Canadian Fish agent in existence [3].

8 What this paper does not establish

- **Nothing about a tuned FishAI.** No parameter was fitted against SESTINA.

- **Nothing about the adaptive or bounded arms** [7, 8].

- **Nothing about the $\sim$16.4-point residual**, which is the largest single term in the deficit and was not measured.

- **Nothing under the corrected `us54` turn-pass rule.** The policy that played these games was tuned inside the engine as it stood before the September 2026 correction [5]. The host engine governed turn flow in every game reported here, so the cells themselves are not invalidated by it; what is unmeasured is whether the corrected policy plays them differently. Re-running the ladder against the corrected arm is the obvious next cell and it has not been run.

- **Nothing about the causal direction of the lock-hold gap** beyond Table 4, per Section 4.1's confound.

9 Conclusion

FishAI plays Canadian Fish at roughly the level of an independently written bot from four releases before the frontier, and it loses to that frontier by 22.92 points. It gets there without a declaration problem: at 98.42% against 98.46% it judges declarations as well as the agent beating it, and at 97.5–98.9% it declares at or above every post-v0.3 bot in that lineage. What it cannot do is establish, quickly, which of its own teammates holds each card of a set its team already owns — and the step in their lineage it fails to clear, v0.3 → v0.4, is exactly the release that added an exact deal posterior. Perfect lock detection is worth +5.75 of the 22.17 points to even, which is a quarter of the gap and a ceiling reached by cheating. The other three quarters are unattributed.

The methodological half is the part this project would keep if it could keep only one. A published number was wrong by 3.44 points because a bridge translated a rules compulsion into a host that does not impose it; the defect was symmetric, so the mirror control could not see it; every corruption counter read zero, because it was mistranslation and not corruption; two adversarial audits passed it; and the one instrument that had already caught it — a protocol counter reading zero where zero is impossible — was filed as a coverage gap. The rules that follow are short. Quote the paired floor, in the unit the pairing is done in. Never cite a mirror cell as evidence of anything. And when a counter reads zero, decide whether zero was possible before deciding it was uninteresting.

References

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[2] FishAI project. WHY-FISHAI-LOSES.md — the account of the cross-play deficit. Repository document, 2026. 139 cells and 166,800 games of re-measurement: §0 the headline table, §2.1 the adapter artifact, §3 the proof-latency channel and its confound, §3.4 the oracle ceiling, §4.1 the lineage placement, §5 what is not the cause, §6.1 the unattributed residual, §8 numbers withdrawn.

[3] `dylann4500`. FishLab — a Canadian Fish engine, bot family and evaluation lab. GitHub repository, `https://github.com/dylann4500/FishLab`. The C++ engine and its identity controls, the SESTINA v1.0 frozen specification and its reported margins, the release history of the v0.2–v0.6 lineage, the guest-bot package protocol, and the pooling caveat on their pathology tool. The repository carries no licence file; nothing from it is reproduced here, and its measured findings are cited as theirs.

[4] FishAI project. MONET.md — the roadmap from Monet v0.1 to v1.0. Repository document, 2026. What the measured ceilings in this paper imply for the 22.92 points still to find, and the decision to reject a `us54` pass rule on the grounds that the defect was in the adapter.

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[6] A. Hsieh. FishAI v0.5: measuring play style without skill in a 54-card Literature variant. FishAI project, 2026.

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[9] A. Hsieh. FishAI v2.0: the three-sided ask. FishAI project, 2026. The defusal term the arm played here carries, and the cross-engine replication of it inside FishLab's TypeScript lab.

[10] FishAI project. ASKING.md — the ask channel and what it publishes. Repository document, 2026. §6 the two corrections that overlap rather than add, cited here for the rule that terms do not sum.